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High-frequency financial data modeling using Hawkes processes

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ABSTRACT

Intraday Value-at-Risk (VaR) is one of the risk measures used by market participants involved in high-frequency trading. High-frequency log-returns feature important kurtosis (fat tails) and volatility clustering (extreme log-returns appear in clusters) that VaR models should take into account. We propose a marked point process model for the excesses of the time series over a high threshold that combines Hawkes processes for the exceedances with a generalized Pareto distribution model for the marks (exceedance sizes). The conditional approach features intraday clustering of extremes and is used to calculate instantaneous conditional VaR. The models are backtested on real data and compared to a competitor approach that proposes a nonparametric extension of the classical peaks-over-threshold method. Maximum likelihood estimation is computationally intensive; we use a differential evolution genetic algorithm to find adequate starting values for the optimization process.

1. Introduction

Volatility structures and regime switches can be observed for wide classes of financial time series as well as at various levels of sampling frequency. Despite a clear conceptual distinction between periods of volatility clustering of a stationary process and changing periods of high/low volatility due to nonstationarity, the nature of the resulting extremes may not always be clearly identified. There is sometimes evidence of an economic regime switch, for instance at the announcement of a merger or takeover, or a management decision concerning interest rates. Our aim is to allow for a wider range of models around the classical Peaks-Over-Threshold (POT) method. The growing availability of tick-by-tick market data from a variety of liquid markets has led to the incorporation of high-frequency data into financial markets and risk management systems. Properly calculated, intraday data can provide significantly improved estimates of risk measures such as Value-at-Risk (VaR) at different time horizons. The pioneering text in the field of high-frequency finance is no doubt Dacorogna et al. (2001). Over the recent years, and this mainly due to the enormous growth in volume of intraday trading, a vast literature has emerged on the subject. Examples which are relevant for our approach are Giot (2005) and Dionne et al. (2009). Our objective is to propose new VaR methodology that characterizes the short-term behavior of tick-by-tick financial time series.

For our purpose, we concentrate on the left tail of the distribution function of the log-returns, hence we are interested in events that lead to extreme losses. In a highly leptokurtic context, the use of an appropriate distribution function from Extreme Value Theory (EVT), like the Generalized Pareto Distribution (GPD), is justified. Classical EVT typically applies to iid random variables. Extensions to more general stationary sequences are possible; see for instance Embrechts et al. (1997). In the case of nonstationarity, one basic solution is to increase the threshold defining the extreme losses until apparent dependence is no longer significant. The resulting drawback is an increase in variance from the reduction of observations. Another simple solution is to first decluster the data and then apply standard methods like the peaks-over-threshold method, leading to a loss of information contained in the clusters. A more sophisticated alternative is the two-stage approach proposed by McNeil and Frey (2000). The first stage models the form of the dependence and the second the amplitude of the declustered residual data. For a literature review of extremes of dependent time series and the resulting challenges, see Chavez-Demoulin and Davison (2012).

An interesting way of dealing with the clustering of events is to allow for a conditional intensity in a marked Poisson process context. The resulting self-exciting Hawkes processes (Hawkes, 1971)

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were first introduced in earthquake occurrence studies by Ogata (1988). They are also known as Epidemic-Type After Shock (ETAS) models (Helmstetter and Sornette, 2002), and are increasingly used in applications in finance. Examples include the modeling of the duration between trades (Bauwens and Hautsch, 2009) or of the arrival process of buy and sell orders (Carlsson et al., 2007). They also can be integrated in a risk management context to estimate conditional value-at-risk (Herrera and Schipp, submitted for publication; Chavez-Demoulin et al., 2005), or to model multivariate financial time series (Embrechts et al., 2011; Errais et al., 2010) in the context of portfolio credit risk. Ait-Sahalia et al. (forthcoming) use Hawkes processes to capture the effect of contagion of jumps in different regions of the world caused by stock market shocks.

The structure of the paper is as follows. In Section 2 we present the complex TAQ database and discuss the need for data pre-processing. Section 3 summarizes the concepts of extreme value theory. In particular, the peaks-over-threshold approach is introduced via a non-homogeneous two-dimensional Poisson process. In Section 4, we explain how to combine the Hawkes process for the counting process of the occurrences and a GPD model for the contributions coming from the marks. This section contains details on parameter estimation using a specific genetic algorithm to maximize a likelihood, together with graphical goodness-of-fit tests. Section 5 presents several resulting models and introduces a self-exciting POT model that links together the counting times and marks. In Section 6, the models are used to calculate conditional risk measures and their applicability is assessed by performing a backtesting procedure.

2. High-frequency financial data: the TAQ database

Electronic exchanges are complex systems that handle a staggering amount of trades each second. Symbols are occasionally simultaneously traded on different exchanges. Mistakes from traders (wrong price, quantity) can sometimes be amended afterwards. Technological progress and regulatory changes also heavily influence the market landscape. For example, shorter trade processing led to the emergence of algorithmic trading where the reaction time is so significant that servers are sometimes located inside the exchange buildings, shaving off a few milliseconds of latency in the process, to gain an edge against competitors; this evolution is referred to as colocation. Short processing times allow traders to exploit new strategies, thus increasing trading frequency and volume although reducing the mean order size. In January 2001, the NYSE changed the minimum tick change from 1/8$ (or 1/16 for stock prices over 10$) to 1/100$. This allowed greater competition between the liquidity providers, thus reducing the spread and, from a statistical point of view, leading to a finer grained distribution of the prices.

The markets are therefore gigantic heterogeneous systems that generate a huge amount of data (multiple terabytes per year for the NYSE alone). We use data provided by the New York Stock Exchange (NYSE) through the Wharton Research Data Services (WRDS). The data are not guaranteed as error-free and are in raw format. The name of the database is the TAQ database for "Trades and Quotes." It contains all the transactions and quotes for nearly all US stocks since 1993.

Table 1
NVDA database structure. The table shows an extract of raw trade data of NVDA (NVIDIA Corporation) from the TAQ; each line is called a tick.

SYMBOL	DATE	TIME	PRICE	SIZE	G127	CORR	COND
NVDA	20060901	9:30:04	28.67	264,509	0	0	–
NVDA	20060901	9:30:04	28.67	264,509	0	2	–
NVDA	20060901	9:30:05	28.65	100	0	0	–
NVDA	20060901	9:30:05	28.65	200	0	0	–
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Table 1 contains an example of raw trade data for NVDA (NVIDIA Corporation) from the TAQ. Each line is called a tick, and the columns are

1. *SYMBOL* or ticker: Denomination of the stock generating the data.
2. *DATE*: Current date of the data.
3. *TIME*: Current time of the data (1 s resolution).
4. *PRICE*: Price of the current trade.
5. *SIZE*: Size of the order.
6. *G127*: Stopped stock trade indicator.
7. *CORR*: Correction indicator, it can take the values:
   - $0 \to 2$ Good trades (0: regular, 1: corrected, 2: symbol correction).
   - $7 \to 9$ Canceled trades due to errors or corrections.
   - $10 \to 12$ Correction instruction (10: cancel, 11: error, 12: correction).
8. *COND*: Condition of sale, with possible values (not exhaustive):
   - A cash-only basis.
   - B average price trade (aggregated trade of same price orders within 60 s).
   - O opening trade reported later.
   - G average price trade not reported with the 90 s exchange limit.

An illustration of the data obtained through the TAQ database on 1 day in January 2006 is given in Fig. 1. The graph highlights occasionally large price spikes, mostly at the beginning of the day. Price spikes might be due to a large order in a period of low volume or to an error. More details about the TAQ database can be found on the NYSE website.

Obtaining good quality high-frequency datasets is clearly more complicated than only processing daily closing prices. Due to the inherent noisiness of financial data and the sheer number of trades settled each day, errors are bound to happen. However, the TAQ database is quite reputable and provides sufficient information so that the data can be pre-processed efficiently.

2.1. Pre-processing

The first pre-processing step is to remove erroneous trades. Our procedure to filter out invalid trades uses four indicators: price, size of order, correction indicator CORR, and condition indicator COND. We chose to apply the following rules proposed in Yan (2007): PRICE > 0, SIZE > 0, CORR < 3, and COND $\neq$ O, Z, B, T, L, G, W, J, K.

A share price of zero or less is obviously not valid. The same is true for the order size. The other two rules are derived from the TAQ database description: A CORR less than 3 indicates a valid trade or a successfully corrected one, and the following cases indicated by COND have been removed: B (average price trade), T (pre/post market trade), Z (sold (out of sequence)), O (opened (late report of opened trade)), L (sold last (late reporting)), G (opening/reopening trade detail), W (average price trade), J (rule 127 trade (NYSE only)), and K (rule 155 trade (AMEX only)). Splits increase liquidity but also influence the perception of the company. Common split ratios are 2–1, 3–1, and 3–2. In the pre-processing step, splits are easily handled by applying the appropriate ratio to prices reported before the split. Information provided from the TAQ database is useful to filter bad data in an accurate way; however, this is not sufficient to obtain the cleanest possible data, as one single trade might cause a spike in the asset price. To remove such data, we use filtering techniques. Signal filtering is an important part of

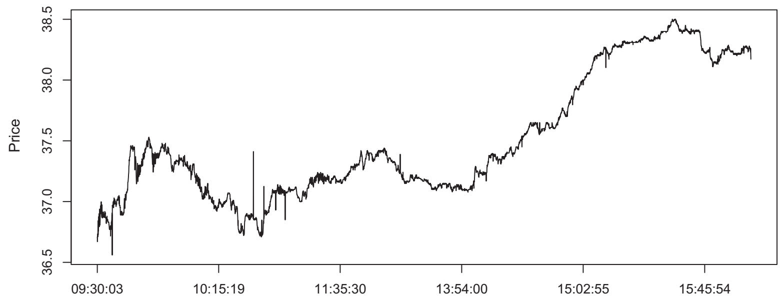

Fig. 1. NVDA stock price January 3, 2006. Raw data from the TAQ database.

data pre-processing in high-frequency data. The simplest techniques commonly applied to high-frequency data are moving average (MA) filters and variance filters. These two techniques are easy to compute, especially the MA filter. The concept of a variance/distributional filter is simply to compare the probability of a measurement under a distribution estimated from a window of past data. If it is lower than a threshold p, the point is removed. The more accurate implementation consists of updating the probability as the window moves on. To provide efficient results, choices relative to the threshold p and the way the probability is computed have to be calibrated on the dataset. Moving average filters are often used because they are simple and effective. The concept of an MA filter is to compare the data with a moving average. If the difference is greater than a threshold parameter, the point is excluded from the dataset. Choice of the window size and the threshold heavily influences the filtered results. The parameters have to be chosen carefully to avoid filtering out good data and keeping too much erroneous data. In this paper, we implement an MA filter and set a window size of 10 and a threshold of 0.1. Using this setup, we removed approximately 0.2% of data.

Even if the TAQ database records the trade to a resolution of a millisecond, the data available through the WRDS platform is provided only with 1-s precision timestamps. Hence, when analyzing data from a liquid stock, there is a high probability that multiple trades occur simultaneously during the same 1-s time interval. There are many approaches to merging these trades, such as averaging or computing the median, but we chose to apply a volume-weighting average, thus giving more weight to large trades. As high-frequency data are irregularly spaced by nature, and as nearly all statistical tools rely on regularly-spaced data, we need some rules to construct a homogeneous time series from the raw data, which is an inhomogeneous series with times t_j and price values $z_j = z(t_j)$. The index j refers to the irregularly spaced sequence of the raw series. The most widely used interpolation methods are the previous point and the linear interpolation. They have useful characteristics and are fast to compute (which turns out to be important when dealing with millions of data points). By using an interpolation method, we construct a homogeneous time series with values at times $t_0 + i\Delta t$, regularly spaced by Δt and rooted at time t_0. The index i then refers to the homogeneous series. Using the notation taken from Dacorogna et al. (2001),

$$j' = \max\{j | t_j \leqslant t_0 + i\Delta t\}, \quad t_{j'} \leqslant t_0 + i\Delta t \leqslant t_{j'+1},$$

so that j' and $j' + 1$ are the two closest measurement indices around the desired interpolation time $t_0 + i\Delta t$. The previous point interpolation is the simplest and fastest method that also has the advantage of preserving causality:

$$z(t_0 + i\Delta t) = z_{j'}.$$

The linear interpolation is defined as follows:

$$z(t_0 + i\Delta t) = z_{j'} + \frac{t_0 + i\Delta t - t_{j'}}{t_{j'+1} - t_{j'}}(z_{j'+1} - z_{j'}).$$

The linear interpolation does not preserve causality, as it uses information not available at the time of interest, so it is not applicable to a real time system. According to Dacorogna et al. (2001), linear interpolation should be favored when studying autocorrelations, and we decide to choose it to pre-process our data.

2.2. Features of high-frequency financial data

The most important characteristics of high-frequency financial data are leptokurtosis (non-normality), autocorrelation between measurements, presence of intraday volatility, and clustering. Most of the biggest losses happen during the opening of the market and show some clustering. The latter is verified by looking at the exponential quantile–quantile plot of the time between consecutive losses, as shown in Fig. 2. The time between negative log-returns above the 95% quantile on 15-min period interpolated NVDA log-returns from 2006 to 2009 does not seem exponentially distributed, and actually we observe higher than expected frequencies for the short distances between extremes. In most applications, independence of widely separated extremes seems reasonable, but extremes often appear in clusters and so display short-range

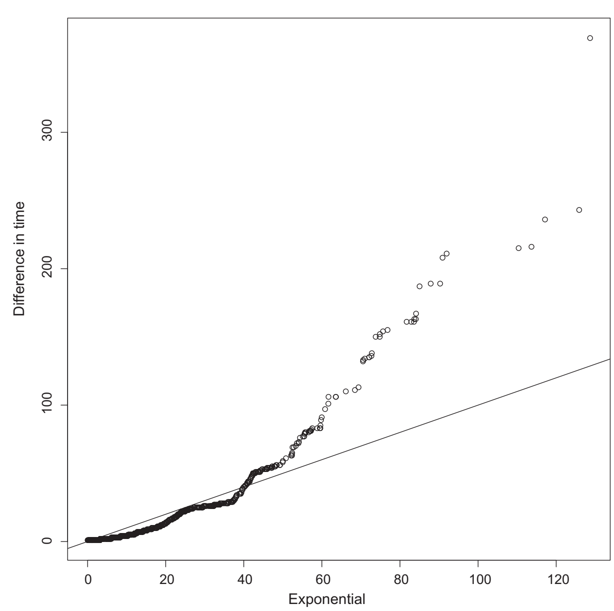

Fig. 2. NVDA log-returns from 2006 to 2009: exponential quantile–quantile plot of the time, in 15-min units, between two successive extreme events.

dependence stemming from serial dependence within the underlying series.

Volatility bursts produce clustered extremes, and it seems unrealistic to assume independent exponential inter-event times and independent mark sizes. These problems are often addressed by the application of a declustering method, which fits the model to cluster maxima only. In a stationary series, it turns out that the distribution of a cluster maximum is the same as the marginal distribution of an exceedance. This raises however the question of how to identify clusters from data. Difficulty in identifying cluster maxima can lead to severe bias in the estimation of parameters. Although appropriate in some applications, declustering is undertaken specifically to avoid modeling the short-term behavior of extremes, which is the central focus of the discussion below. This paper introduces a marked point process model for the behavior of clusters of high-frequency extremes, by combining a self-exciting process for the threshold exceedance times with a GPD model for the threshold excess sizes. The form of the process allows realistic models in which the recent past affects the current intensity of arrival of events more than does the distant past, and the form allows the intensity of arrival of events to depend on the marks.

3. Extreme value theory

3.1. The block maxima method

Consider the maximum of a sequence of independent and identically distributed random variables $Z_1,\ldots,Z_q$ from a continuous distribution function F. Denote by z_0 the upper support point of F. For any fixed $z < z_0$ we have

$$P(\max(Z_1,\ldots,Z_q) \leqslant z) = P(Z_i \leqslant z, i = 1,\ldots,q) = F^q(z),$$

which tends to 0 as $q \to \infty$. To obtain a non-degenerate limiting distribution for the maximum, the Z_i must be rescaled by sequences $\{a_q > 0\}$ and $\{b_q\}$ of normalizing constants leading to $Y_q = a_q^{-1}\{\max(Z_1,\ldots,Z_q) - b_q\}$. As $q \to \infty$, a possible limiting distribution of Y_q is such that $P(Y_q \leqslant y) = F^q(b_q + a_q y)$ has a limit. It can be shown that the latter expression, rewritten as $\left[1 - \frac{q\{1-F(b_q+a_q y)\}}{q}\right]^q$, has a limit if and only if $\lim_{q\to\infty} q\{1 - F(b_q + a_q y)\}$ exists. If so, the only possible limit is

$$\lim_{q\to\infty} q\{1 - F(b_q + a_q y)\} = \begin{cases} \left(1 + \xi\frac{y-\mu}{\psi}\right)_+^{-1/\xi}, & \xi \neq 0, \\ \exp\{-(y-\mu)/\psi\}, & \xi = 0, \end{cases} \tag{1}$$

where $(x)_+ = x$ if $x > 0$ and 0 otherwise, with $-\infty < \xi,\ \mu < \infty$, and $\psi > 0$. In theory, this requires that F satisfies some properties of regular variation (de Haan, 1970), but in the vast majority of applications, simple conditions are sufficient. This leads to the remarkable result that given suitable sequences $\{a_q\}$ and $\{b_q\}$, the non-degenerate limiting distribution must be a generalized extreme value (GEV) distribution

$$H(y;\mu,\psi,\xi) = \exp\left\{-\left(1 + \xi\frac{y-\mu}{\psi}\right)_+^{-1/\xi}\right\}.$$

The parameters $-\infty < \mu < \infty$ and $\psi > 0$ are respectively the location and scale parameters of the distribution. The parameter $-\infty < \xi < \infty$ controls the shape of the distribution.

There are three particular forms of H depending on the value of ξ:

- $\xi = 0$ (Fisher–Tippett Type I or Gumbel), which corresponds to
$$H(y;\mu,\psi,0) = \exp\left\{-\exp\left(-\frac{y-\mu}{\psi}\right)\right\}, \quad -\infty < y < \infty;$$
- $\xi > 0$ (Fisher–Tippett Type II or Fréchet), which has a finite lower support point; and
- $\xi < 0$ (Fisher–Tippett Type III or Weibull), which has a finite upper support point.

A standard application of this limiting result is to fit the GEV distribution to a series of block maxima, taking q to be the number of observations in a block (month, week, minute, second, etc.), and to use the fitted distribution to estimate the $1/p$-block return level, that is, the level exceeded once on average every $1/p$ block $(0 < p < 1)$. The quantity $1/p$ is called the return period, and the usual return level estimate is

$$y_p = \begin{cases} \mu - \psi[1 - \{-\log(1-p)\}^{-\xi}]/\xi, & \xi \neq 0, \\ \mu - \psi\log\{-\log(1-p)\}, & \xi = 0, \end{cases} \tag{2}$$

where parameters are replaced by their estimates.

Standard errors can be calculated from the observed information matrix, and the classical asymptotic properties of maximum likelihood estimators hold throughout the range $-1/2 < \xi < \infty$ (Smith, 1985). In most financial contexts, maximum likelihood estimation techniques may be applied safely. The confidence intervals for parameters and quantiles that are based on asymptotic theory might be poor when few data are available. The confidence intervals for return level estimates are often skewed and therefore poorly approximated by a Normal distribution. Alternative approaches include the profile likelihood method (Davison and Smith, 1990) where the problem is reparameterized in terms of $(y_{(1-p)},\ \mu,\ \xi)$, treating ξ and μ as nuisance parameters.

3.2. Peaks-Over-Threshold (POT)

Suppose we observe points in a time interval $[0,t_0]$. Let $Z_1,\ldots,Z_{\lceil qt_0\rceil}$ be an independent and identically distributed series from an unknown distribution F, and consider the two-dimensional pattern, subset of the plane $S = [0,t_0] \times [u,\infty)$, with points $(t,\ y)$ at coordinates $\left(j/q, a_q^{-1}(Z_j - b_q)\right)$, $j = 1,\ldots,\lceil qt_0\rceil$. The idea is to characterize the behavior of the points at high levels. Providing that a_q and b_q are such that a limiting distribution for the maximum Y_q exists as $q \to \infty$, the pattern of points in the region of the form $A = [t_1,t_2] \times [y,\infty)$ with y sufficiently large will converge to an inhomogeneous Poisson process. A basic property of such a process is that the number of points in any nonoverlapping regions of the set $[0,t_0] \times [u,\infty)$ are independent. The event $\left\{a_q^{-1}(Z_j - b_q) > y\right\}$ implies that the number of points in A can be expressed as

$$\Delta_q(A) = \sum_{j=\lceil qt_1\rceil}^{\lfloor qt_2\rfloor} I(Z_j > b_q + a_q y),$$

with $0 \leqslant t_1 < t_2 \leqslant t_0$. As the Z_j with distribution function F are independent and identically distributed, the intensity measure of the limiting process $\Delta_q(A)$ as $q \to \infty$ is binomially distributed with parameters $(n,\ p)$ with $n = \lceil qt_1\rceil - \lfloor qt_2\rfloor + 1$ and $p = 1 - F(b_q + a_q y)$. Hence the Poisson limit for the binomial distribution combined with (1) yields

$$P\{\Delta(A) = n\} = \frac{\Delta(A)^n}{n!}\exp\{-\Delta(A)\}, \quad n = 0,1,2,\ldots,$$

with

$$\Delta(A) = \begin{cases} (t_2 - t_1)\left(1 + \xi\frac{y-\mu}{\psi}\right)_+^{-1/\xi}, & \xi \neq 0, \\ (t_2 - t_1)\exp\{-(y-\mu)/\psi\}, & \xi = 0, \end{cases} \tag{3}$$

with $0 \leqslant t_1 < t_2 \leqslant t_0$ and $y > u$. Noting that $\Delta(A)$ has the form $\Delta_1([t_1, t_2])\Delta_2([y, \infty))$, the process can be split into two independent Poisson processes with corresponding intensity measures Δ_1 and Δ_2. The first describes the random times at which $Z_j > u$, and the second gives their rescaled sizes $a_q^{-1}(Z_j - b_q)$. This construction forms the basis of the threshold method described below. The foregoing result gives a characterization for all observations that are extreme in the sense of having exceeded a high threshold u. The exceedance times occur according to a homogeneous Poisson process with intensity $\tau = \{1 + \xi(u - \mu)/\psi\}_+^{-1/\xi}$, and their sizes follow an inhomogeneous Poisson process with intensity derived from

$$\lambda_u = -\frac{d\Delta\{[y,\infty)\}}{dy} = \begin{cases} \frac{1}{\psi}\left(1 + \xi\frac{y-\mu}{\psi}\right)_+^{-1/\xi-1}, & \xi \neq 0, \\ \frac{1}{\psi}\exp\left\{-\frac{(y-\mu)}{\psi}\right\}, & \xi = 0. \end{cases} \tag{4}$$

Note that this intensity does not depend on t but does depend on y; hence, the two-dimensional Poisson process is non-homogeneous. It follows from (3) that for any $Z \geqslant u$, the implied one-dimensional process of exceedances of the level y is a homogeneous Poisson process with rate $\tau(y) := -\ln H(y; \mu, \psi, \xi)$. Now we consider the excess amounts over the threshold u. The tail of the excess distribution function over the threshold u, denoted $\overline{F}_u(y)$, can be calculated as the ratio of the rates of exceeding levels $u + y$ and u. We obtain

$$\overline{F}_u(y) = \frac{\tau(u+y)}{\tau(u)} = \left(1 + \frac{\xi y}{\psi + \xi(u-\mu)}\right)^{-1/\xi} = \overline{G}_{\xi,\sigma}(y),$$

for a positive scaling $\sigma = \psi + \xi(u - \mu)$. This is precisely the tail of the Generalized Pareto Distribution model for excesses over the threshold u. The characterization described above can be used to show that the number of exceedances over threshold u has a Poisson distribution with mean $t_0\tau$ and that conditional on n exceedances; their sizes $W_j = Z_j - u$ are a random sample of size n from the Generalized Pareto Distribution

$$G_{\xi,\sigma}(w) = \begin{cases} 1 - (1 + \xi w/\sigma)_+^{-1/\xi}, & \xi \neq 0, \\ 1 - \exp(-w/\sigma), & \xi = 0. \end{cases} \tag{5}$$

As $\xi \to 0$, $G_{\xi,\sigma}(w)$ tends to the exponential distribution with mean σ. Eq. (5) can be used as the basis for a likelihood for σ and ξ which is

$$l(\sigma, \xi) \doteq -n\log\sigma - (1 + 1/\xi)\sum_{j=1}^{n}\log(1 + \xi w_j/\sigma),$$

and the overall loglikelihood is then

$$l(\lambda, \sigma, \xi) \doteq n\log\tau - t_0\tau - n\log\sigma - (1 + 1/\xi)\sum_{j=1}^{n}\log(1 + \xi w_j/\sigma).$$

Estimation of the parameters ξ and σ of a generalized Pareto random variable is non-regular if $\xi < -1/2$ (Davison, 1984a,b; Smith, 1985). For $\xi > 1$, it has an infinite mean. Typically, for financial time series, ξ is strictly positive.

There are simpler ways of getting the same parameter estimates by using a reparametrization of the POT model in terms of $\tau := \tau(u) = -\ln H(u; \mu, \psi, \xi)$, the rate of the one-dimensional Poisson process of exceedances of the level u, and $\sigma = \psi + \xi(u - \mu)$, the scaling parameter of the implied GPD for the excess losses over u. The intensity (4) can then be rewritten as

$$\lambda(y) = \lambda(t, y) = \frac{\tau}{\sigma}\left(1 + \xi\frac{y-\mu}{\sigma}\right)^{-1/\xi-1}, \tag{6}$$

where $\xi \in \mathbb{R}$ and $\tau, \sigma > 0$.

The choice of the threshold u is important. Smith (1987) proposes a graphic technique to assess the fit of the model: a "mean residual life plot" in which the mean excess over a threshold u is plotted against u, for a wide range of values u. This suggestion is based on the fact that if Z is generalized Pareto with parameters σ and ξ, then, when $\xi < 1$, $u \geqslant 0$ and $\sigma + u\xi > 0$, we have

$$E(Z - u|Z \geqslant u) = \frac{\sigma + u\xi}{1 - \xi}.$$

If the generalized Pareto assumption is correct, the plot should be linear in u with intercept $\sigma(1 - \xi)$ and slope $\xi/(1 - \xi)$. In practice, the plot tends to be very irregular at its right-hand end, when u is large and the number of exceedances over u small. Elsewhere, however, it can be a very useful tool to check the appropriateness of the generalized Pareto distribution. Based on an extensive comparative simulation study, Chavez-Demoulin (1999) recommends choosing a threshold such that about between 5% and 10% of the data are excesses. Herrera and Schipp (submitted for publication) propose a sensitivity analysis based on a mean squared error to assess the stability of the VaR among different thresholds. In our empirical study, we use between 5% and 8% of the largest losses. Using the POT approach, the $1/p$-period return level (2) becomes

$$y_p = u + \frac{\sigma}{\xi}\{(\tau/(1-p))^{\xi} - 1\}.$$

In finance, this quantity is the value-at-risk, where the return period $1/p$ is replaced by the confidence level $1 - p$, for p close to one.

4. Hawkes process representation

Under the assumptions required by the standard POT approach, the loss generating process can be modeled with a marked Poisson process of constant intensity loss arrivals with GPD marks (sizes). If the loss arrivals are indeed driven by a Poisson process, time between arrivals should follow an exponential distribution. As already mentioned in Section 2, Fig. 2, this is not the case for intraday data. Using the same example of Bayer daily shares used by Chavez-Demoulin et al. (2005), in the top panels of Fig. 3, the presence of clusters of extremes over the threshold (which corresponds to the 95% quantile) is clearly shown both by the plot of the exceedances above the threshold (left panel) and by an exponential quantile plot (right panel) that does not produce a straight line. In this case, the standard POT model cannot be applied without violating its assumptions. The lower panels show the same two representations but for the declustered data above the same threshold. The declustering uses a "runs" method described in Embrechts et al. (1997) (Embrechts et al., 1997, pp. 419–423). A simple rule is applied to form the clusters: extremes separated by more than the runs method parameter r are considered to belong to a different cluster. This method systematically defines the location of each cluster and picks the largest. We use $r = 30$ for the Bayer data, and the declustering clearly allows the application of the standard POT model. One drawback of the declustering is that the difficulty in identifying cluster maxima (choice of the threshold and of r) can lead to severe bias (Fawcett and Walshaw, 2007). This bias can be reduced by using all exceedances to estimate the GPD parameters, but then the standard errors must be modified to allow for the dependence. Another drawback of the declustering is that even if the model fits, it becomes difficult to interpret the results, as we modified the original data. Removing clusters erases a large quantity of information about the underlying process.

In this section, we discuss a Poisson process that allows a type of dependence between the marks, the Hawkes process. This family of models was introduced in the seminal paper by Hawkes (1971). Historically, the main field of application for this family of models has been seismology, which models the likelihood of earthquakes knowing that the probability increases in relation to the amplitude of past events (Ogata, 1988). In a financial context, Hawkes processes have been applied recently to model risk in

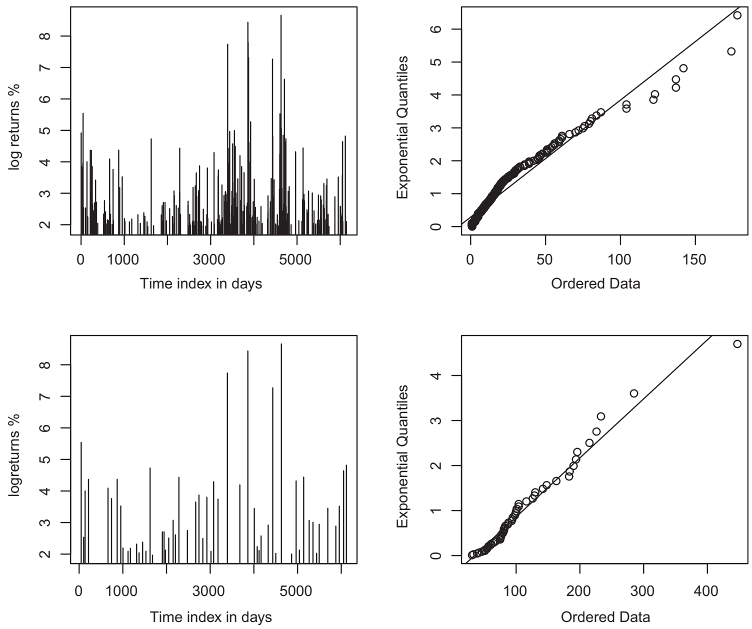

Fig. 3. Bayer negative daily log-returns. Nondeclustered negative log-returns above the 95% quantile and exponential quantile–quantile plot of the time in days between two successive such events (upper right). The lower panels are the same plots but after declustering using a runs method with $r = 30$.

financial assets (see for instance Chavez-Demoulin et al. (2005), Ait-Sahalia et al. (forthcoming), McNeil et al. (2005)), in credit risk (Giesecke and Zhu, forthcoming), or to predict buy and sell order arrival times (Carlsson et al., 2007).

Following the notation from Chavez-Demoulin et al. (2005), we consider an event process $(T_i, W_i), i \in \mathbb{Z}$ where T_i defines the times and W_i the marks, which when observed over the period $(0, t_0]$ give data $(t_1, w_1), \ldots, (t_n, w_n)$. Let $\mathscr{H}_t$ be the sigma algebra generated by the process, that is, the entire history of the marks and their occurrence up to time t. The joint density of the data seen over $(0, t_0]$ can be written

$$\prod_{i=1}^{n} f_{T_i, W_i | \mathscr{H}_{t_{i-1}}}(t_i, w_i | \mathscr{H}_{t_{i-1}}) P(T_{n+1} > t_0 | \mathscr{H}_{t_n}),$$

where $\prod_{i=1}^{n} f_{T_i, W_i | \mathscr{H}_{t_{i-1}}}(t_i, w_i | \mathscr{H}_{t_{i-1}})$ defines the joint density of $(T_1, W_1), \ldots, (T_n, W_n)$ given $\mathscr{H}_0$, and $P(T_{n+1} > t_0 | \mathscr{H}_{t_n})$ ensures that there are exactly n data observed over the period $(0, t_0]$. Assuming independence of times T_i and marks W_i the conditional density function is

$$f_{T_i, W_i | \mathscr{H}_{t_{i-1}}}(t_i, w_i | \mathscr{H}_{t_{i-1}}) = f_{T_i | \mathscr{H}_{t_{i-1}}}(t_i | \mathscr{H}_{t_{i-1}}) f_{W_i | \mathscr{H}_{t_{i-1}}}(w_i | \mathscr{H}_{t_{i-1}}),$$

leading to the log-likelihood

$$\ell = \left(\sum_{i=1}^{n} \log f_{T_i | \mathscr{H}_{i-1}}(t_i) + \log P(T_{n+1} > t_0 | \mathscr{H}_{t_n}) \right) + \sum_{i=1}^{n} \log f_{W_i | \mathscr{H}_{t_{i-1}}}(w_i | \mathscr{H}_{t_{i-1}}) = l_A + l_B. \quad (7)$$

Combining separately a Hawkes process for the counting process of the times (part A of (7)) and a GPD model for the contributions from the marks (part B), the likelihood becomes

$$\ell = \log \left\{ \prod_{i=1}^{n} \lambda(t_i | \mathscr{H}_{t_i}) \exp\left(-\int_0^{t_0} \lambda(u | \mathscr{H}_u) du \right) \right\} - n \log \sigma - (1 + 1/\xi) \sum_{i=1}^{n} \log(1 + \xi w_i / \sigma), \quad (8)$$

with $\lambda(t | \mathscr{H}_t)$ being the conditional intensity. For details on the theory of counting processes, see Cox and Isham (1980). We chose the following general form with a constant loading μ_0:

$$\lambda(t | \mathscr{H}_t) = \mu_0 + \sum_{j: t_j < t} \omega(t - t_j; w_j),$$

where $\mu_0 > 0$ and the self-exciting function $\omega(s)$ has to be positive when $s \geqslant 0$ and 0 elsewhere. For example, the conditional intensity model proposed by Ogata is

$$\lambda(t | \mathscr{H}_t) = \mu_0 + \psi \sum_{j: t_j < t} \frac{e^{\delta w_j}}{(t - t_j + \gamma)^{1+\rho}}, \quad (9)$$

where $\mu_0, \psi, \delta, \gamma, \rho > 0$. This model belongs to the epidemic-type after shock family, and its properties have been thoroughly studied (Helmstetter and Sornette, 2002). In the financial high-frequency context, it is natural to also let ω depend on the marks w_j. The impact of large losses on the intensity of the process is driven exponentially by the amplitude of the losses and decreases in relation to the distance from the event. The short-term clustering is adequately taken care of by the large intensity increase following extreme losses.

Estimation of the resulting likelihood can be computationally intensive. We use a differential evolution genetic algorithm (Storn and Price, 1997) to find adequate starting values for the classical optimization routine. Genetic algorithms are very useful techniques when dealing with optimization problems in finance (for details see http://www.icsi.berkeley.edu/~storn/code.html). Differential evolution is used for multidimensional real-valued functions. In contrast with standard optimization methods (such as gradient descent and quasi-Newton methods), it does not use the gradient and therefore does not require the optimization problem to be differentiable. It can be used for optimization problems that change over time and are not continuous. The optimization process creates a population of candidate solutions and makes them reproduce and mutate through multiple generations. The individual which has the best "fitness" across the final population is considered as the solution. The optimization problem actually provides a measure of quality at any candidate solution instead of looking at its gradient.

There are many different techniques to judge the goodness-of-fit of a Hawkes process (Ogata, 1988). If the model is correct, the conditional intensity should be able to predict the arrival of events closely. Considering the random time change, $\Lambda_{\mathscr{H}_t}(t) = \int_0^t \lambda(u|\mathscr{H}_u)du$, the cumulative number of events realized during the estimation period should follow a Poisson process of unit intensity (Papangelou, 1972). The lower panel in Fig. 5 shows the estimated cumulative number of events $\widehat{\Lambda}_{\mathscr{H}_t}(t)$ for the 342 exceedances of the NVDA 2008 stock of 15-min interpolated negative log-returns above the 95% quantile, using the ETAS model (9) with $\rho = 0$ (values of the estimated parameters are given in Section 5). The resulting curve lies within the 95% and 99% Kolmogorov–Smirnov bounds, lending support to the fitted model.

5. Applications to high-frequency financial data

We consider the NVDA prices (upper panel) and negative log-returns (lower panel) for the years from 2006 to 2008 represented in Fig. 4. The dataset contains in total 20,384 15-min linearly interpolated returns. Volatility structures and regime switches can be observed at various levels of sampling frequency, leading to non-stationarity. There is hence a need to allow for a wider spectrum of models around the classical POT method. We apply the ETAS model (9) to the negative log-return data, considering a high threshold corresponding to the 95% quantile. The estimation is done by maximizing the likelihood (8) using the optimizing procedure explained in Section 4. The estimated parameters are

$$\widehat{\mu_0} = 0.009(0.004), \quad \hat{\psi} = 1(1), \ \hat{\gamma} = 26(15), \ \hat{\delta} = 0.06(0.02), \ \hat{\rho} = 0.33(0.17).$$

A likelihood ratio test shows that all the parameters are significant at the 5% level except ρ (p-value = 0.07). The simplest model with $\rho = 0$ in (9) provides the estimated parameters

$$\widehat{\mu_0} = 0.007(0.004), \quad \hat{\psi} = 0.14(0.02), \ \hat{\gamma} = 6(3), \ \hat{\delta} = 0.06(0.02).$$

All the parameters are significant at a 5% level, except μ_0; therefore the process seems to be purely driven by the self-exciting part. Price movement is driven by recent events, and there is no evidence for a deterministic trend. The estimated intensity function for 2008 obtained from the model, here $\rho = 0$, is represented in Fig. 5, upper panel. The high volatility period starting from September is highlighted, with an increase in the probability of an extreme loss against the rest of the year; recall that Lehman Brothers filed for Chapter 11 bankruptcy protection on September 15, 2008. The lower panel shows the cumulative events number $\widehat{\Lambda}_{\mathscr{H}_t}(t)$ (solid line) and the two-sided 95% and 99% confidence limits (dashed diagonal), based on the Kolmogorov–Smirnov statistic; the straight diagonal line represents the perfect fit of the model. The cumulative intensity lies within overall confidence limits and gives no evidence against the model.

We consider a further model discussed in McNeil et al. (2005) consisting of a Hawkes process with exponential decay:

$$\lambda(t|\mathscr{H}_t) = \mu_0 + \psi \sum_{j:t_j<t} \exp\{\delta w_j - \gamma(t - t_j)\}. \tag{10}$$

The estimated parameters are

$$\hat{\mu}_0 = 0.050(0.003) \quad \hat{\psi} = 3 \times 10^{-6}(2 \times 10^{-5}) \ \hat{\gamma} = 0.006(0.002) \ \hat{\delta} = 0.1(0.6).$$

The only significant parameter at the 5% level is the location parameter μ_0. Not surprisingly, $\hat{\mu}_0$ is equal to the predetermined proportion of extremes. The results are identical to those obtained by a standard POT method. The backtesting analysis (Section 6) shows that the standard POT model underestimates the VaR calculation.

A possible generalization of the POT model can be obtained by incorporating a self-exciting component where marks have a GPD, but are predictable. This so called self-exciting POT model is discussed in McNeil et al. (2005). The exceedances are conditionally GPD, given the exceedance history up to the time of the mark, with a scaling parameter that depends on that history. Such a model can be justified easily, for example, by the volatility pattern that is observed throughout the day. As previously stated, most extreme

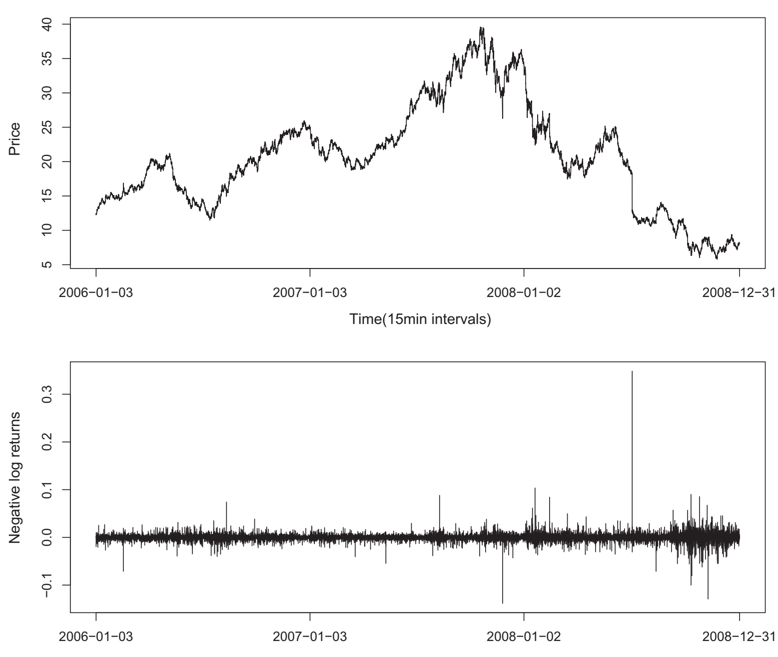

Fig. 4. NVDA stock 15-min interpolated price (upper panel) and negative log-returns (lower panel) from 2006 to 2008.

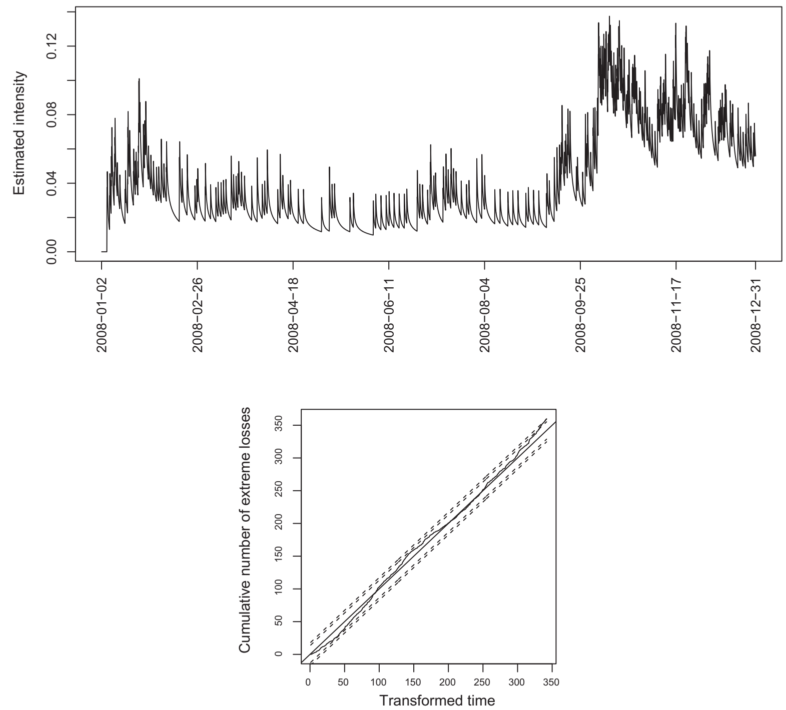

Fig. 5. NVDA 2008 stock 15-min interpolated negative log-returns: estimated intensity function for 2008 obtained from the model (9) with $\rho = 0$ (upper panel) and estimated cumulative events number $\widehat{\Lambda}_{\mathscr{H}_t}(t)$ (lower panel). The dashed diagonal lines are the two-sided 95% and 99% confidence limits based on the Kolmogorov–Smirnov statistic.

price movements take place during the opening of the exchange and, to a smaller extent, during the closing. Describing this phenomenon as well as possible is thus of great importance when modeling financial data. In a period of high excitement, both the temporal intensity of occurrence and magnitude of the exceedances increase. The model uses the representation of the standard POT model given by (6) where the exceedances of the threshold u occur as a homogeneous Poisson process with rate τ and excesses are GPD with parameters ξ and σ. The self-exciting POT model with predictable marks is obtained by generalizing (6) to get

$$\lambda^{\star}(t,y) = \frac{\tau + \alpha_1 v(t)}{\sigma + \alpha_2 v(t)} \left(1 + \xi \frac{y-u}{\sigma + \alpha_2 v(t)}\right)^{-1/\xi - 1}, \tag{11}$$

where $v(t) = \sum_{j:t_j<t} \omega(t-t_j)$ uses a self-exciting function $\omega(s)$. Empirical analysis shows that it is reasonable to treat the shape parameter ξ as a constant. The two previous forms of the self-exciting process driven by the ETAS model (9) for ω and the exponential decay Hawkes process (10) are still justified. For the latter, all the parameters are significant at a 5% level and so differ from the standard POT model. Fig. 6 shows the threshold exceedances, corresponding to 5% of the largest values for the 2008 NVDA data (upper panel). The estimated intensity of exceeding the threshold equals $\int_y^\infty \lambda^{\star}(t,s)ds$, where v uses the form (9) (middle panel), and the estimated conditional GPD mean $(\sigma + \alpha_2 v(t))/(1-\xi)$ (bottom panel). In the self-exciting POT model, both the estimated intensity and GPD mean follow the path of the function $v(t)$.

6. Conditional risk measures and backtesting

In practice, the purpose of fitting our model is to estimate the conditional p%VaR of the predictive distribution for the return over the next period using the data observed up to time t:

$$z_p^{t+1} = F^{-1}_{Z_{t+1}|\mathscr{H}_t}(p),$$

with

$$\begin{aligned}\overline{F}_{Z_{t+1}|\mathscr{H}_t}(z) &= P(Z_{t+1} > z|\mathscr{H}_t) = P(Z_{t+1} - u > z|Z_{t+1} > u, \mathscr{H}_t) \\ &\quad \times P(Z_{t+1} > u|\mathscr{H}_t).\end{aligned}$$

In the case of the ETAS models, $P(Z_{t+1} > u|\mathscr{H}_t)$ represents the conditional probability of an event in $(t, t+1]$, that is,

$$1 - P\{N(t,t+1) = 0|\mathscr{H}_t\} = 1 - \exp\left(-\int_t^{t+1} \lambda(u|\mathscr{H}_u)du\right).$$

The term $P(Z_{T+1} - u > z|Z_{t+1} > u, \mathscr{H}_t)$ is estimated using the GPD model with parameters σ and ξ. The required quantile z_p^{t+1} is then the solution of the equation:

$$P(Z_{T+1} > z_p^{t+1}|\mathscr{H}_t) = 1 - p,$$

with parameters replaced by their maximum likelihood estimates. From the estimated conditional VaR, it is straightforward to get an estimation of the conditional Expected Shortfall (ES) as

$$\mathrm{ES}_p^{t+1} = \frac{z_p^{t+1}}{1-\xi} + \frac{\sigma - \xi u}{1-\xi}.$$

The self-exciting POT model (11) can also be used to calculate a conditional VaR,

$$z_p^{t+1} = u + \frac{\sigma + \alpha_2 v(t+)}{\xi} \left\{ \left(\frac{1-p}{\tau + \alpha_1 v(t+)} \right)^{-\xi} - 1 \right\},$$

where $t+$ denotes the time point just after t. The associated conditional Expected Shortfall (ES) can then be calculated by observing that the conditional distribution of the excess losses above z_p^{t+1}, given information up to time t, is GPD with shape parameter ξ and scaling parameter given by $\sigma + \alpha_2 v(t+) + \xi(z_p^{t+1} - u)$.

Fig. 7 shows the estimated VaR (red curve) and ES (blue curve) at the 95% level for the next 15-min period from end of July to end of December 2008, for the ETAS model (9) with $\rho = 0$. The green

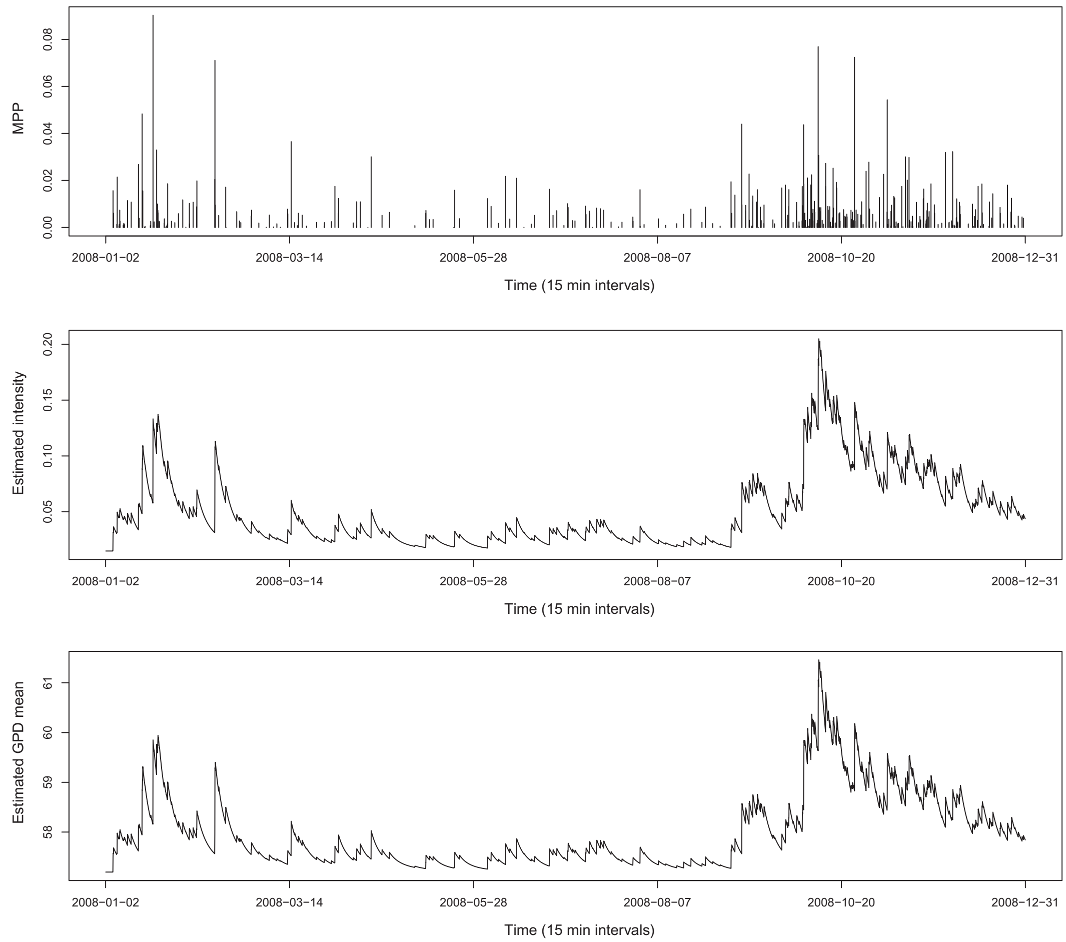

Fig. 6. NVDA 2008 stock 15-min interpolated negative log-returns: the upper panel shows the exceedances over the threshold (95% quantile). The peak value higher than 0.3 shown in Fig. 4 (lower panel) has been removed in order to better appreciate the exceedances fluctuations. The middle panel represents the estimated intensity of exceeding the threshold in a self-exciting POT model with ω of the form (9), and the bottom panel shows the estimated mean of the conditional GPD of the excess above the threshold.

curve shows the 95% VaR estimates obtained from the classical POT model. The VaR estimates are out-of-sample (forecasts) so that the 95% VaR estimates obtained from the classical POT model vary in time due to the incorporation of new information at each time. A backtesting study shows that the resulting VaR estimates tend to underestimate the real VaR values because the time where the series shows low regimes of volatility (as shown in the period before September 2008) has the same weight in the calculation as the latest period of high volatility.

The upper panel in Fig. 8 shows the estimated VaR for the next 15-min at the 95% and 99% levels from end of July to end of December 2008, computed from the self-exciting POT model (11) with $v(t) = \sum_{j:t_j<t} \frac{e^{\delta w_j}}{(t-t_j+\gamma)^{1+\rho}}$. Procedures based on residuals are useful to assess the goodness-of-fit of the model. One possible graphical diagnostic is based on the result that the residuals

$$R_j = \log\{1 + W_j\hat{\xi}/\hat{\sigma}\}/\hat{\xi}, \quad j = 1, \ldots, T,$$

are distributed approximately as independent unit exponential variables when the model is correct. Looking at the lower panel in Fig. 8, it seems that the dependent form for the GPD scale parameter is convincing.

We compare the different models presented in Section 5 in terms of their backtesting performance on samples from the test period of 2000 and 4000 data points. The test period of length 2000 is randomly chosen over the entire period from 2006 to 2008, and the sequence of 4000 observations corresponds to the second half of 2008. We use the sample of 4000 observations to backtest the models for the VaR at high levels (99% and 99.9%). We calculate the VaR from the different models at every 15-min interval of the test period. We compare our different Hawkes process models with the classical POT method (considering stationary iid series) and with an alternative approach also based on POT proposed by Chavez-Demoulin et al. (forthcoming). Their so-called NPOT method consists of a nonparametric extension of the classical POT method to fit the time-varying volatility in situations where the stationarity assumption is maybe violated by erratic changes of regime for daily data. Applied to the 15-min interpolated negative log-returns of NVDA data, Fig. 9 shows the 95% VaR and ES estimated from end of July to end of December 2008. The curves react to abrupt changes of volatility in more "squared" moves than the Hawkes process, which produces a more smoothed aftershock decreasing curve.

To backtest the model, consider the sample of size T of observed value $\{z_{t+1}\}_{t=0}^{T-1}$ and estimated VaR out-of-sample interval forecasts $\left[\hat{z}_p^{t+1}, \infty\right)$ at a level p calculated at time t using a certain window size of past observations. At each time $t+1$, we introduce the

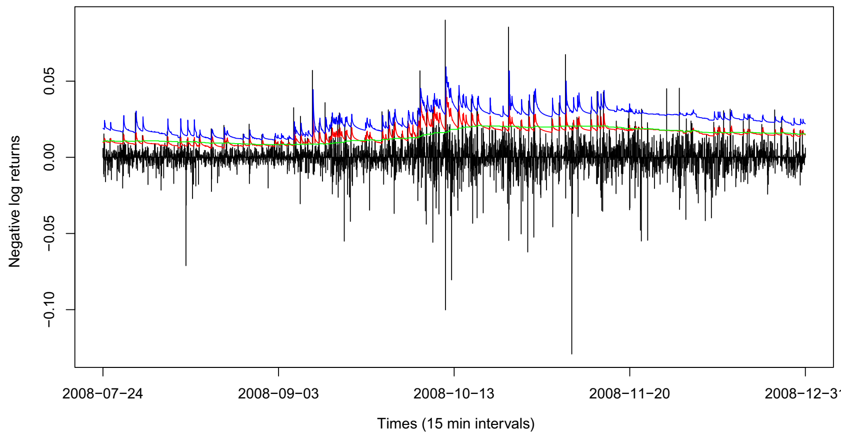

Fig. 7. NVDA 2008 stock 15-min interpolated negative log-returns from end of July to end of December 2008: estimated VaR (red curve) and ES (blue curve) at the 95% level for the next 15-min period computed from the ETAS model (9) with ρ set to zero. The green curve shows the estimated VaR obtained from the classical POT method.

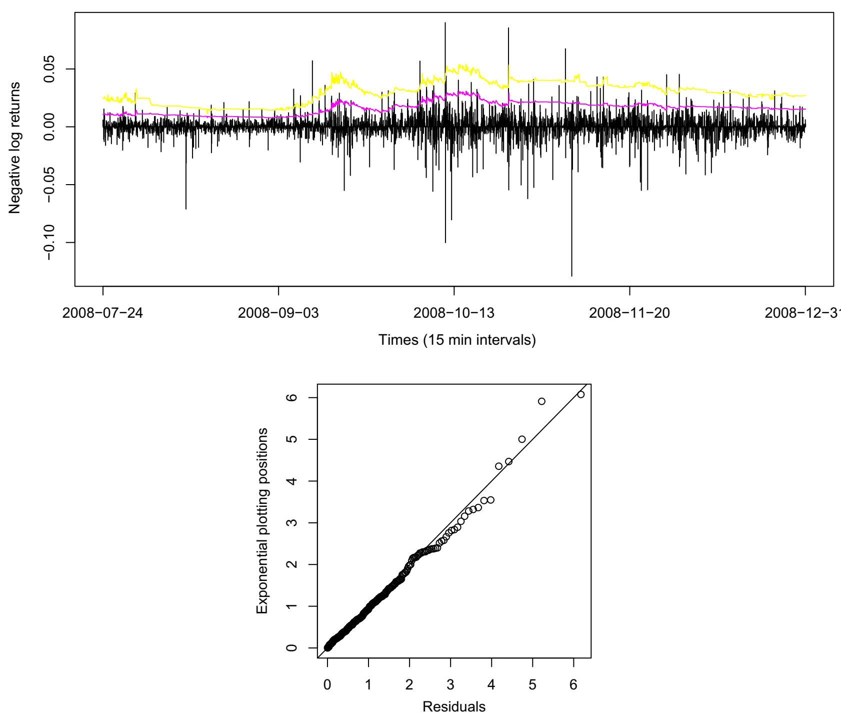

Fig. 8. NVDA 2008 stock 15-min interpolated negative log-returns: estimated VaR at the 95% and 99% levels (upper panel) for the next 15-min period from end of July to end of December 2008 computed from the self-exciting POT model (11) with $v(t)$ of the form (9). The lower panel shows an exponential probability plot of the residuals using the GPD scale dependent parameter.

violation indicator variable I_{t+1}, which compares the p%VaR, $\hat{z}_p^{t+1}$, with the observed value at time $t + 1$, z_{t+1}:

$$I_{t+1} = \begin{cases} 1, & \text{if} \quad z_{t+1} \in \left[\hat{z}_p^{t+1}, \infty\right) \\ 0, & \text{otherwise} \end{cases}.$$

Standard evaluation of interval forecasts proceeds by simply comparing the nominal coverage $\hat{p} = \sum_{t=0}^{T-1} I_{t+1}/T$ to the true coverage p. Assuming that the indicators are independent, the total number of violations is binomially distributed under the model. This can be tested (Christoffersen, 1998) by means of a likelihood ratio test,

$$LR_{uc} = 2\log\left[L(\hat{p}; I_1, \ldots, I_T)/L(p; I_1, \ldots, I_T)\right] \overset{\text{asy}}{\sim} \chi_1^2,$$

where L is the binomial likelihood and $\overset{\text{asy}}{\sim}$ means asymptotically distributed. The drawback of this unconditional coverage testing method is that it fails to properly characterize the behavior of the model in the presence of clustering. It may happen that, although the number of violations is correct, they might occur in clusters. To take into account that possibility, we also perform the test of independence and the test of conditional coverage suggested in Christoffersen (1998). The first tests the independence assumption of the indicators, and the second tests for both independence and correct coverage leading to a complete test of correct conditional

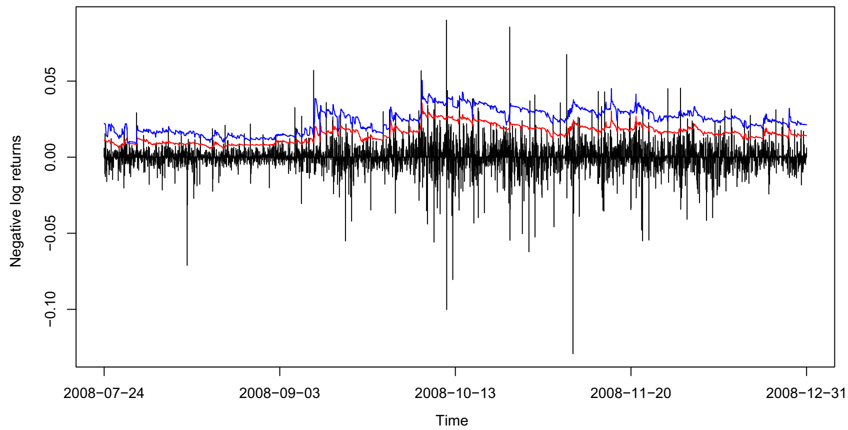

Fig. 9. NVDA 2008 stock 15-min interpolated negative log-returns from end of July to end of December 2008: estimated VaR and ES at the 95% level for the next 15-min computed from the NPOT method.

coverage, without making any hypothesis about the underlying true conditional distribution. The test of independence is performed by modeling the number of violations I_{t+1} as a binary first-order Markov chain with transition probability matrix

$$\Pi = \begin{bmatrix} 1-\pi_{01} & \pi_{01} \\ 1-\pi_{11} & \pi_{11} \end{bmatrix},$$

where $\pi_{ij} = P(I_{t+1} = j | I_t = i)$. Conditional on the first indicator, the approximate joint likelihood for this model is

$$L(\pi_1; I_2, \ldots, I_T | I_1) = (1-\pi_{01})^{n_{00}} \pi_{01}^{n_{01}} (1-\pi_{11})^{n_{10}} \pi_{11}^{n_{11}},$$

where n_{ij} denotes the number of transitions from state i to state j. The maximum likelihood estimators are

$$\hat{\pi}_{01} = \frac{n_{01}}{n_{00}+n_{01}}, \quad \hat{\pi}_{11} = \frac{n_{11}}{n_{10}+n_{11}}.$$

We can estimate a first-order Markov chain model from our observed sequence of indicators $\{I_{t+1}\}$ by noting that $\pi_{01} = \pi_{11} = \pi_2$ corresponds to independence. The conditional likelihood under the null hypothesis is

$$L(\pi_2; I_2, \ldots, I_T | I_1) = (1-\pi_2)^{(n_{00}+n_{10})} \pi_2^{(n_{01}+n_{11})},$$

with maximum likelihood estimate $\hat{\pi}_2 = (n_{01}+n_{11})/\ (n_{00}+n_{10}+n_{01}+n_{11})$. The likelihood ratio test is

$$LR_{ind} = 2\log\left[L(\pi_1; I_2, \ldots, I_T | I_1)/L(\pi_2; I_2, \ldots, I_T | I_1)\right] \overset{asy}{\sim} \chi^2_1.$$

This test does not depend on the true coverage p but provides the independence test of our hypothesis. To test correct conditional coverage, we jointly test for independence and correct probability parameter p. Christoffersen (1998) suggests combining the unconditional coverage test and the independence test using

Table 2
NVDA 2008 stock 15-min interpolated negative log-returns: Number of violations, unconditional coverage test, independence test, and conditional coverage test results at levels p = 95% and p = 99% and p = 99.9% for the different models: *Classical POT* is the standard method for iid data; *ETAS* is the model for exceedance times with $\nu(t)$ of the form (9); *Exp* is the model for exceedance times with $\nu(t)$ of the form (10); *ETAS POT* represents the self-exciting POT model with $\nu(t)$ of the form (9); *Exp POT* represents the self-exciting POT model with $\nu(t)$ of the form (10); and *NPOT* is the competitor model. The results are the number of violations observed over the sample and the p-values. The models have been backtested on a sample of size T = 2000 for the 95% level and T = 4000 for the 99% and 99.9% levels.

	Number of violations	Unconditional	Independence	Conditional
0.95 Level	Expected = 100	T = 2000		
Classical POT	337	$<10^{-4}$	NA	NA
ETAS	106	0.53	0.075	0.16
Exp	106	0.53	0.075	0.16
ETAS POT	96	0.71	0.054	0.14
Exp POT	99	0.95	0.031	0.099
NPOT	110	0.30	0.22	0.26
0.99 Level	Expected = 40	T = 4000		
Classical POT	58	0.0066	0.00016	$<10^{-4}$
ETAS	37	0.69	0.35	0.57
Exp	37	0.69	0.35	0.57
ETAS POT	35	0.47	0.43	0.52
Exp POT	35	0.47	0.43	0.52
NPOT	45	0.42	0.53	0.60
0.999 Level	Expected = 4	T = 4000		
Classical POT	25	$<10^{-4}$	NA	NA
ETAS	7	0.12	0.87	0.39
Exp	7	0.12	0.87	0.39
ETAS POT	8	0.069	0.85	0.20
Exp POT	7	0.12	0.87	0.39
NPOT	11	0.0028	0.80	0.015

$$LR_{cc} = LR_{uc} + LR_{ind} \overset{\text{asy}}{\sim} \chi^2_2.$$

Table 2 gives the number of expected and observed violations and the p-values for the different models at levels $p = 95\%$, $p = 99\%$ and $p = 99.9\%$ on the sample test periods of length $T = 2000$ for the 95% level and $T = 4000$ for the 99% and 99.9% levels. All the tests show a poor VaR estimation obtained from the classical POT model, which tends to underestimate the VaR. According to the results, there is no evidence against correct estimation of the VaR at the three levels for all the Hawkes models. Interestingly, the unconditional coverage testing provides lower p-values at the 99.9% level than the conditional test. This is due to the fact that the p-values for the independence test are rather high, meaning that the violations do not appear to be clustered. For the lower levels (95% and 99% with higher expected and observed number of violations), the conditional test takes into account the occasional dependence of the violations, leading to (still acceptable) lower p-values than the unconditional test, which ignores that violations can appear in clusters. Compared to NPOT, the methods are competitive, though there is a slight underestimation of the VaR at 99.9% for the NPOT model in this case.

The results of our Hawkes process are therefore comparable to those obtained in Chavez-Demoulin et al. (2005) for the daily frequency data and using the ETAS model (9) with ρ significantly equal to zero. In their paper, a further simulation study confirms the adequacy of the Hawkes process on a daily basis.

As expected, the performance of the standard POT lies in the stationarity of the series, and the high volatility of the 2008 test period leads to an excessive number of violations. According to the backtesting, conditional VaRs seem to be correctly estimated for all the models with no real difference in performance among them, and in light of these studies, it seems that our Hawkes processes for the excesses over high threshold provide an adequate model for high-frequency extremal returns showing strong clustering and high volatility.

7. Discussion

High-frequency trading currently represents about 70% of the trading volume in the US equity market, and there is a pressing need to explore and study possible models for high-frequency extremal returns. In this paper, we offer a family of Hawkes processes that appropriately capture the volatility clustering behavior of the intraday extremal returns. According to the backtesting, the different models are competitive, though further studies on specific periods of much lower volatility show that the self-exciting POT models perform rather better than the ETAS models applied only to the counting process of the times. This is certainly due to the fact that a unique self-exciting function is used for both the frequency and magnitude of the POT model, adapting quickly to change of regimes. The Hawkes process provides a suitable estimation of high-quantile based risk measures (VaR, ES) for financial time series. Moreover, credible regions can be derived, which would provide financial analysts with a valuable measure of short-term uncertainty. Methods presented in this paper can be extended to multivariate time series data as proposed in Embrechts et al. (2011) or to portfolios for which high quantiles need to be tracked.

Acknowledgments

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